

Detection Sensitivity Analysis of Otsu Thresholding for Active Region Recovery in Aditya-L1 SUIF NB03 Imagery

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ISA Summer Training Program 2026

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Abstract—This work presents a detection sensitivity analysis for an Otsu-thresholding-based active region detection pipeline applied to Aditya-L1 SUIF NB03 (Mg II k, 2796 Å) Level-1 imagery from May 9, 2026. The threshold value was varied from 0.4 to 0.65 in the interval of 0.05 which is almost symmetrical to Otsu threshold value 0.5341. The number of unique NOAA active regions were recovered at each threshold value and cross validated against the Heliophysics Event Knowledgebase (HEK) catalog. A plot which shows the comparison between unique NOAA AR and its corresponding threshold value is also made to understand the trend. The results are consistent with four of the six tested regions detecting six unique NOAA regions. Seven counts were detected for threshold value of 0.45 and five counts at threshold value of 0.6. This demonstrates the reliability of Otsu-based thresholding for disk-central regions and threshold sensitivity for regions which are close to the solar limb.

Index Terms—Otsu thresholding, solar active regions, Aditya-L1, SUIF, sensitivity analysis, image segmentation

I. INTRODUCTION

In the earlier part of the project (Part 1), we established a pipeline which detects active region from Aditya L1 SUIF's imagery using Otsu thresholding to compute a intensity threshold. The analysis of image taken on 9 May 2026 (07:07:21 UTC) by SUIF gave us 61 active region blobs and 7 unique NOAA active regions and the Otsu threshold was found to be 0.5341. It also included AR14436 which subsequently gave a M5.7 flare and CME on May 10.

Since Otsu's threshold is a statistically estimated quantity rather than a fixed physical constant, in this work we investigate how sensitive the pipeline's final output is to small changes around this baseline threshold value. A pipeline whose active-region recovery count changes drastically under small threshold shifts would indicate fragile detection logic, whereas a pipeline whose output remains stable indicates a methodologically reliable approach.

II. RESEARCH QUESTION

How is the number of unique NOAA regions affected by varying threshold values around Otsu's threshold on Aditya L1's SUIF imagery taken on 9 May 2026 at 07:09:21 UTC?

III. METHODOLOGY

A. Reuse of Prior Pipeline Stages

For this analysis we would be using the array `arr_2d_masked` which is a flat-field corrected, percentile

normalized and circular disk masked array in part 1. This is the same array on which Otsu thresholding was also applied. We have imported that array directly into this project's notebook for saving time and avoiding repetition of steps. The corresponding `sunpy.map.Map` object is also retained in this notebook, as it provides the World Coordinate System (WCS) information required for pixel-to-world coordinate transformation during NOAA cross-matching.

B. Threshold Grid

Six threshold values were generated using `numpy.arange(0.40, 0.66, 0.05)`, yielding `{0.40, 0.45, 0.50, 0.55, 0.60, 0.65}`, which as can be seen are symmetrically located to Otsu's threshold of 0.5341.

C. NOAA Catalog

The HEK NOAA active region catalog for May 9, 2026 was queried once, before the loop for sensitivity, as the catalog of active regions present on this or any date is independent of the image-processing threshold applied.

D. Per-Threshold Processing

For each threshold value in the list, the following steps were performed in the sequence mentioned below:

- 1) **Thresholding:** Each threshold in the list was applied to `arr_2d_masked` to get its corresponding binary output.
- 2) **Labeling:** Connected components were labeled using `skimage.measure.label`, and region properties (area, centroid) found via `regionprops`.
- 3) **Adaptive artifact removal:** The former pipeline (Part 1) used a fixed area cutoff (`max_threshold_area = 10000`) to remove the solar limb artifact. But this same approach is not useful in the present case as the maximum blob area ranged from $\approx 1.4 \times 10^6$ pixels at threshold 0.40 to $\approx 5.2 \times 10^3$ pixels at threshold 0.6. Clearly, a fixed cutoff of 10,000 would not be able to identify the artifact at high thresholds and will not work to the purpose at lower thresholds. To solve this, an adaptive cutoff approach was implemented which was computed afresh at each threshold value. This approach is useful in a way that it changes as per the need of the threshold ensuring that the artifacts are removed

from every image. The threshold is calculated as 99th percentile of that iteration's blob-area distribution using `numpy.percentile(areas, 99)`.

- 4) **Fixed off-disk artifact removal:** This was done to remove the off-disk distortions present in the western and southern hemispheres of the sun (as indicated by PRADAN) as much as possible.
- 5) **NOAA cross-matching:** The cleaned mask was then labeled again, and each blob centroid converted from pixel coordinates to Heliographic Stonyhurst coordinates via `pixel_to_world` followed by a frame transform. A blob is considered matched to a NOAA active region if its angular separation from the HEK-reported position was within 5° which is a fair approximation.
- 6) **Recording:** Threshold value, total blob count, and the set of uniquely matched NOAA AR numbers were recorded after each iteration.

IV. RESULTS

TABLE I
UNIQUE NOAA ACTIVE REGIONS RECOVERED AS A FUNCTION OF OTSU THRESHOLD VALUE.

Threshold	Blob Count	Unique NOAA ARs
0.40	1397	6
0.45	3680	7
0.50	11037	6
0.55	4565	6
0.60	2178	6
0.65	1669	6

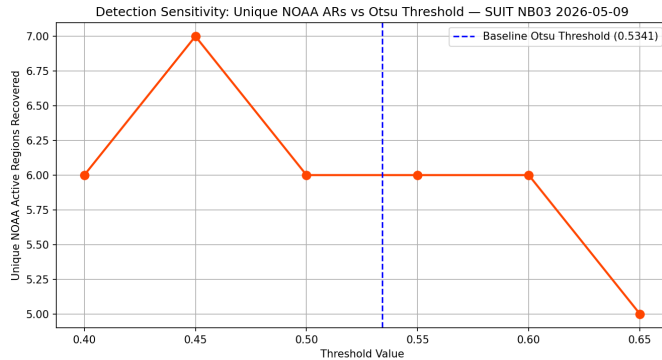


Fig. 1. Unique NOAA active regions recovered versus Otsu threshold value, with the baseline threshold (0.5341) marked as a vertical dashed line.

The number of unique NOAA regions identified peaked at threshold value 0.45 with 7 unique identified regions and remained stable at 6 across four of the six tested threshold values. 5 unique regions were identified at threshold value of 0.65, the lowest in all of our tested thresholds.

V. DISCUSSION

A. Threshold-Dependence of AR14436

The elevated count at threshold 0.45 is attributable to AR14436 which is the region responsible for the M5.7 flare and CME observed on May 10, 2026. AR14436 was matched

only at threshold 0.45, and absent at all other tested values, including 0.50, the nearest tested value to the baseline of 0.5341.

This is because of AR14336's location on 9 May 2026. At that date, it is located at ≈ -81 Heliographic Stonyhurst longitude which means that it lies near the solar limb on the western side. At lower thresholds, more pixels satisfy the binarization criterion, and AR14336's fragmented, signal most probably merges with limb, which is subsequently removed in morphological cleanup. At higher threshold values, fewer pixels will be detected as active and hence due to their small area they will be filtered out from minimum area condition or they could also be in the region where coordinate generation is not happening near the disk edge. Threshold 0.45 falls within a sweet narrow region and it is where AR14436 is detected.

From this analysis it should not be interpreted that 6 is the true active-region count near baseline threshold of 0.5341. Rather, the result tells us that the detection of regions near limb is highly sensitive to threshold values. A coarser threshold interval can fail to sample that window.

B. Stability of Core Detections

The six regions which are found consistently across nearly all tested thresholds (AR14429, AR14431, AR14432, AR14433, AR14434, AR14435) indicate that these areas were bright enough and disk-central to be detected in spite of changes in threshold values. This tells us that Otsu-based thresholding is reliable for regions not present near limb.

C. Limitations

- The adaptive 99th-percentile cleanup generalizes and automates the original pipeline's limb-removal logic across multiple threshold values, but it does not contain its manual removal of on-disk horizontal and vertical patches, which relied on exact area values specific to the baseline threshold and do not generalize across a threshold sweep.
- Coordinate transformation warnings (non-rotation transform warnings, NaN angular separations) were observed for blobs near to the solar limb. They were more frequent at lower thresholds where even more blobs survive. But this is an expected consequence of converting 2D helioprojective coordinates to Heliographic Stonyhurst (HGS) coordinates near solar disk. This doesn't affect the validity of matches for on-disk regions.
- The threshold grid (0.40–0.65, step 0.05) does not include the exact baseline (0.5341); results at threshold 0.50, the nearest tested value, should not be treated as equivalent to the baseline result.

VI. CONCLUSION

The pipeline's active region detection count remains largely stable (6 unique NOAA ARs) across a ± 0.10 – 0.15 range around the Otsu threshold baseline, with the notable exception of the threshold value 0.45, which detects the region AR14436, its detectability is threshold-sensitive within a narrow window. This demonstrates the reliability of Otsu-based thresholding

for disk central regions and a limitation for regions closer to solar limb.